

Joaquin Alec P. Hao

Full-Stack Software Engineer • Next.js / React / TypeScript • Supabase

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Professional Summary

Computer Engineering graduate from Jose Rizal University, **Cum Laude**, and a full-stack engineer who ships production software solo. Built and deployed VoyageAI—a collaborative AI travel-planning SaaS—end-to-end with Next.js 16, React 19, TypeScript, Supabase (PostGIS, RLS, Realtime), and the Vercel AI SDK, at zero infrastructure cost. Strong across the stack: REST APIs, relational and realtime databases, role-based access control, and AI-assisted delivery with Claude Code. **Open to remote opportunities.**

Education

Jose Rizal University

Mandaluyong City, Philippines

Bachelor of Science in Computer Engineering • 2022 – 2026

Graduated **Cum Laude**

Thesis: Design and Development of an Intelligent Robotic Assistant for Campus Information Management

Experience

IT Group Inc.

February 2026 – April 2026

Position Title: Student Intern

- Administered NetSuite ERP user roles, permissions, and sandbox environments for employee access management.
- Built and executed SuiteQL queries and saved searches to extract, filter, and validate employee and partner datasets.
- Cleaned, merged, and reconciled 3,000+ customer records and 4,000+ item records to maintain data integrity prior to system migration.
- Imported and updated employee records including job titles, name formatting, and supervisor IDs into NetSuite.
- Developed an ISMS Data Governance Handbook covering 7 company systems: NetSuite, Celigo, Zoho CRM, Zoho Projects, Analytics Plus, Service Desk Plus, and BigQuery.
- Created a finalized ISMS documentation collection covering Governance, Access & Administration, Operations, and Evidence.
- Prepared a formal user license proposal identifying current vs. proposed role changes for business development positions.
- Developed a NetSuite training plan including a syllabus, slide decks, and a user guide for system onboarding sessions.

PLDT-Smart Innovation Generation Pitch Competition

November 2024 – September 2025

Position Title: Participant • **Giga 25 Finalist**

- Proposed a mobile application and website platform focusing on job application and services for persons with disabilities.
- Created and conceptualised a concept paper and wireframe.
- Lead developer for backend and database for the mobile application.
- Constructed and connected the database to the system using Firebase and Android Studio.
- Main compiler for the mobile application, connected both the frontend and backend.

Byte Forward Hackathon 2025

August 2025

Position Title: Participant

- Proposed a mobile application focusing on assisting non-tech-savvy people in getting started with digital services within 24 hours.
- Designed and developed the backend portion of the application.
- Implemented AI chatbot feature.

Amazon Web Services Learning Clubs – JRU Chapter

March 2025 – Present

Position Title: The Secretariat (Co-founding Executive)

- Co-founding executive of the AWS Learning Clubs chapter at Jose Rizal University.
- Hosted and attended seminars, workshops, and events exploring topics and opportunities in AWS cloud computing.
- Handled the organization's documents and finances.

Analytics & Artificial Intelligence Association of The Philippines

February 2025 – January 2026

Position Title: Data Engineer Intern

- Managed and backed up the files of the organization.
- Cleaned CRM data to be migrated to the new database.
- Attended conventions, increasing knowledge on Analytics and Artificial Intelligence.

Projects

VoyageAI — Collaborative AI Travel Planner

May 2026

- Solo-built a production-grade SaaS for collaborative group travel using Next.js 16, React 19, TypeScript, Supabase, and Gemini AI—deployed live on Vercel at zero infrastructure cost.
- Engineered an AI optimization pipeline: DBSCAN proximity clustering → nearest-neighbor TSP scheduling → four-axis fairness scoring (satisfaction, Jain's fairness index, efficiency, weather-fit) → human-reviewable changes applied transactionally.
- Built a grounded AI assistant with Vercel AI SDK v6 tool calls into live trip data (preferences, itinerary, weather, budget)—supports Gemini and Groq across 8 free models; never invents itinerary items.
- Implemented three-layer security: PostGIS Row-Level Security (data visibility), server-action role checks (permissions), and UI gating (affordances)—access control written once, enforced everywhere.
- Delivered full production deployment using free-tier-only providers: Gemini, MapTiler, Nominatim, OpenWeather, Frankfurter, Supabase, and Vercel.

Pokémon TCG Online Binder Manager

December 2025

- Developed a full-stack web application for organizing and managing Pokémon Trading Card collections with 1,000+ card capacity per digital binder.
- Integrated Pokémon TCG REST API with advanced filtering (set, type, rarity, language) and pagination for card search across 10,000+ cards.
- Migrated cloud database from Firebase Firestore to Supabase for improved query performance and scalability.
- Built drag-and-drop interface for card management and customizable UI themes with image upload functionality.
- Optimized API queries reducing payload size by 90% and improving search performance from 3–5s to sub-1s response times.
- Developed end-to-end with Claude Code as the primary development workflow—a production example of AI-assisted engineering.

THRIVE — Mobile Job Application Platform

June 2025

- Developed an inclusive mobile job application platform tailored for Persons with Disabilities (PWDs) using Android Studio and Firebase.
- Designed and implemented the app's Firebase Real-time Database structure to manage user profiles, job listings, and application flows.
- Built core backend logic to support user authentication, data storage, and the job application process.
- Assisted in frontend and UI/UX design.

Intelligent Robotic Assistant for Campus Information Management

2025 – 2026

B.S. Computer Engineering Thesis — Commissioned by the Dean, JRU

- Built a RAG pipeline using a multi-LLM architecture enabling the robot to answer campus queries from an uploaded knowledge base via natural conversation.
- Developed an emotion detection module (dataset gathering, cleaning, model training) that adapts responses to the user's detected facial expression.
- Integrated LiDAR, ultrasonic, and IR sensors for autonomous navigation; led cross-team software–hardware integration across 9 members.
- Built a web-based admin panel for remote startup, knowledge-base management, and hardware health monitoring.

Thesis Library Website

March 2024

- Developed a web-based thesis library for Jose Rizal University using HTML, PHP, and MySQL to help students and faculty browse, search, and access academic research efficiently.
- Led database design, integrated phpMyAdmin for connectivity, and implemented PDF upload and download functionality.
- Led project planning, system architecture, and collaborated with school faculty to meet university requirements for publication.

Skills & Interests

AI & Developer Tools: Claude Code (AI-assisted development), Vercel AI SDK v6, Gemini AI, Groq, Prompt Engineering, RAG pipelines

Frontend: HTML, CSS, TypeScript, ReactJS, Next.js 16, Tailwind CSS

Backend: Java, PHP, Python, Next.js App Router, REST APIs

Database & Cloud: MySQL, PostgreSQL, Supabase (PostGIS • RLS • Realtime), Firebase, AWS, Vercel

ERP & Data Engineering: NetSuite, SuiteQL, Data Governance, ISMS Documentation, BigQuery

Hardware & Embedded: Embedded Systems & IoT, Circuit Design, Robotics, Arduino, ESP32

Interests: Exploring emerging technologies and innovations, gadget tinkering, circuit designing, and robotics development.

Certifications

- AWS Certified Cloud Practitioner
- AWS Cloud Quest: Cloud Practitioner
- DataCamp Certified Data Engineer Associate
- CompTIA ITF+ Certified